

# Science

Nature Walks & Scouting

General Science

Natural History

Nature Notebook





## About the Course

In level 2, learners are developing familiarity and friendship with Creation through a broader scope of Things, more challenging text, and slightly more abstract concepts.

This course includes the following topic(s): General Science: Grade 2, Natural History: Grade 2, Nature Notebook: Grade 2, Nature Walks & Scouting: Grades 1-8

### About Nature Walks & Scouting: Grades 1-8

Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

### About General Science: Grade 2

A primarily practical course, students explore the idea of biome, first their own and then others. They also 'meet' some less familiar creatures from their own, such as insect larva and mammals, through picture books. Coordinating afternoon activities are provided in Outdoor Work.

### About Natural History: Grade 2

Students learn to observe a variety of creatures through their Nature Reader, including insects, insect larvae, mammals, and some aquatic species. Coordinating afternoon activities are provided in Outdoor Work.

### About Nature Notebook: Grade 2

Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

### Science: Grade 2

#### The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Form 1 learners in grades 1-3 are at the beginning of this relationship: developing familiarity and friendship with Creation. Their sense of community is primarily focused on their immediate environment at this stage, but they are curious about others. The familiarity with foundational knowledge grown and nurtured in Form 1 prepares them for greater curiosity and personal interest in Form 2.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just 'feels right.' Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Literature Grades 1-3).



## Placement & Combining Tips

### Science: Grade 2

For second-grade students or possibly hungry first-graders and third-graders taking their time. If combining grades is desirable, it is recommended that the combined grades begin at the appropriate level and move forward together.

### Nature Walks & Scouting: Grades 1-8

Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.

### Nature Notebook: Grade 2

Learners may be combined and share prompts or follow their own interest. Support through varied media, scribing observations, or notebooking in tandem.



## Scheduling

| GRADE | SCHEDULE INFO.                                             | BOOKS                                                                                                                |
|-------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1-8   | Nature Walks & Scouting: Grades 1-8<br>1 time/week 30 min+ |                                                                                                                      |
| 2     | General Science: Grade 2<br>1 time/week 15 min             | A Walk in the...<br>Desert Giant<br>Mama Built a Little Nest<br>Monarch and Milkweed<br>The Secret Life of the Skunk |
| 2     | Natural History: Grade 2<br>1 time/week 10 min             | Walks with a Naturalist: Living Nature Reader Volume 2                                                               |
| 2     | Nature Notebook: Grade 2<br>1+ time/week 10 min+           |                                                                                                                      |

### Sample Weekly View

| Day 1                                                | Day 2 | Day 3                                                           | Day 4 | Day 5 |
|------------------------------------------------------|-------|-----------------------------------------------------------------|-------|-------|
| <b>Science: Grade 2</b>                              |       |                                                                 |       |       |
| General Science: Grade 2<br>Nature Notebook: Grade 2 |       | Natural History: Grade 2<br>Nature Walks & Scouting: Grades 1-8 |       |       |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

### Science: Grade 2

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

### Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved

around in the schedule to incorporate or substitute Natural History Club outings.

☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

### Special Topics & Field Trips

Science: Grade 2

- ☐ Learn a few of your local species on these special topics using the internet, field guides, or your local nature preserve, conservancy, or state/national park.
- ☐ Note whether there are particular locations nearby for the student to easily notice the special topics on walks or during afternoon occupations.

General Science: Grade 2

- Term 1: insects, especially ants and butterflies, flowers, trip to a natural history museum or butterfly house
- Term 2: birds and trees
- Term 3: animals, tree buds, trip to a pond and a creek, hopefully with wet/soft ground to observe animal tracks

Natural History: Grade 2

- Term 1: insects, especially larvae, and trees
- Term 2: animal sign, twigs, marine creatures
- Term 3: insects, birds, spring ephemerals

### Term Prep & Teacher Tips

Science: Grade 2

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term:

General Science: Grade 2

- ☐ Term 1: butterflies (Monarchs) p.305-310 and wildflowers, milkweed p.491-495 or aster p.506-509
- ☐ Term 2: trees p.618-626, bird nests p.46, and/or choose 1-2 local birds from p.50-143
- ☐ Term 3: skunks p.245-247 and/or choose one additional mammal in p.215-266

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- any wildflower seeds or butterfly host plants (Term 1)
- pot/planter with soil or place in the ground to plant with a trowel
- optional: rock

Natural History: Grade 2

- ☐ Term 1: insects p.294-300, trees p.618-623
- ☐ Term 2: tree winter work p.624, choose 1-2 mammals from p.214-293
- ☐ Term 3: wildflower intro and 1-2 spring ephemerals from p.460-487, aquatic insects and dragonflies p.400-408

### Reminders

General Science: Grade 2

☐ Raising caterpillars is planned for General Science in Term 1, so make a note in your calendar to order them at the appropriate time and be flexible with the schedule to accommodate their development and release according to the instruction that comes with them.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Science: Grade 2

#### General Science: Grade 2



A Walk in the...



Desert Giant



Mama Built a Little Nest



Monarch and Milkweed



The Secret Life of the Skunk

#### Natural History: Grade 2



Walks with a Naturalist: Living Nature Reader Volume 2



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

#### Nature Walks & Scouting: Grades 1-8



Wild Bird Seed



Bird Feeder



Hand Lens



Small Collection Containers



Small Basin



Dip Net



Student Microscope

#### General Science: Grade 2



Butterfly Garden with Voucher to Obtain Larva



Household Items - General Science: Grade 2



Various Seashells



## Quick Links

### Science: Grade 2

- ∞ [Outdoor Work](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [SkyView Lite for iOS](#)

Click THIS text  
or scan the QR  
code for links.



# Science: Grade 2

## How To Teach



### Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap. The lesson plans often help with this.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



### Complete

- Read, or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their nature journal or some other notebook/paper available in case they need to draw or diagram during the lesson.
- When reading to pre-readers, point to the words as you read, even if it seems like they aren't paying attention.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, or beyond.



### Narrate

- Process the ideas of the lesson by retelling the events in sequence, describing a scene or setting, explaining a concept, etc.
- Learners may use words, pictures, Legos, dolls, etc., to process and convey ideas in the manner most natural to them.
- Teachers may take turns to model.



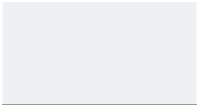
### Discuss

- Consider together any thoughts, confusion, or concerns about the passage.
- Questions/topics for further discussion are often provided in the lesson plans to help. There are no right or wrong answers to these. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.



### Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner interest.

- 
- These can also be viewed as alternative ways to engage.
-





## Term 1

### WEEK 1 15m General Science: Grade 2 - Lesson 1

#### *Looking for Ants Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### → INTRO

Today, you will look for ants!

#### → OBSERVE

Some good places to look are under small rocks, around logs, and in cracks. Use a stick to safely lift rocks. If you find ants, take some time to observe them. Use a magnifier if you have one. How many sections of the body can you see? From which do the legs come? Describe its antennae, jaws, and eyes. If you give it a drop of honey or a few crumbs, what does it do? If you find something else that you want to observe, you may take the time to do so. Spend about 10 minutes looking for ants.

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Did you find ants? If so, describe their habitat. How does it provide food and shelter for them?

#### ★ TEACHER TIP

If students seem anxious or unsure about their nature journal, teachers can encourage them by modeling, showing them many different examples, journaling in tandem, or serving as a scribe.

#### ★ TEACHER TIP

If students find ants, this is a great opportunity for an object lesson. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.

#### ★ TEACHER NOTE

If you haven't ordered your caterpillars yet, don't forget!

### WEEK 1 10m Natural History: Grade 2 - Lesson 1

#### *Night Visitors*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

What do you recall about insects? Briefly share in word or picture. Our naturalist friend, Melissa, is going to introduce us to some of the creatures she sees after dark. Look at the picture on p.1. Have you ever seen such a brightly painted moth? Who else do you think we might meet after dark?

#### → READ, NARRATE, & DISCUSS

p.1-5 "One summer night" - "after dark?"

Read the definitions and notes at the end of each reading, as desired.

- What have you seen or heard when you've been outside after dark? Tell what you have noticed.

#### → SUPPLEMENTAL

∞ Video Link: 7 Spectacular Moths in Slow Motion (5:23)

∞ Video Link: Why moths are obsessed with lamps (2:06)

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

#### ★ TEACHER NOTE

The second video is from an evolutionary perspective. Teachers might choose to allow it to pass by the way, provide additional commentary, or skip this one.

#### • OUTDOOR WORK

Weekly walks and daily outdoor



## Term 1

### WEEK 2 15m General Science: Grade 2 - Lesson 2

*Insects and Plants*

Materials: Monarch and Milkweed

PREP: Read Teacher Tip

#### → INTRO

Look at the picture on the cover. What kinds of butterflies have you noticed before? What did they look like? Where were they?

#### → READ, NARRATE, & DISCUSS

Monarch and Milkweed (whole book)

- The meadow where the flowers grow might be a good habitat for a butterfly, but they might also fly to other habitats to find flowers, as well. A 'biome' is all of the living things that live together in a community in neighboring habitats. Can you think of some different habitats in our biome, where we live, where butterflies might be able to find food and shelter?

time are vital! Resources in Quick Links.

#### ★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 2 10m Natural History: Grade 2 - Lesson 2

*Caterpillar Hunting*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

Tell me about some of the creatures we met last week. All of those creatures were adults, the grown up animal. Today we're going to look for some baby insects, called caterpillars. Have you ever found a caterpillar? Briefly tell what you noticed.

#### → READ, NARRATE, & DISCUSS

p.7-9 "One afternoon" - "a caterpillar?"

- The caterpillars that Melissa found hatched from eggs and would someday turn into adult moths. Draw each of these stages on separate index cards or pieces of paper. Can you put them in order? Which comes first, middle, or last in the life of the insect? (egg -> caterpillar/larva -> adult/moth) Keep these in your book for next week!

#### → SUPPLEMENTAL

∞ Video Link: Luna Moth (1:39)

∞ Video Link: Ceanothus Silk Moth (0:38), a West Coast moth sometimes confused for a luna

#### ★ TEACHER TIP

If students need support drawing or diagramming, the teacher may do any activities with them in tandem or act as their scribe.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 1

### WEEK 3 15m General Science: Grade 2 - Lesson 3

#### *Observing Caterpillars Activity*

Materials: Nature Journal, caterpillars

PREP: Read Teacher Tip

#### → INTRO

Let's take a close look at our caterpillars today.

#### → OBSERVE

Spend about 10 minutes observing your caterpillars in their habitat today and plan to check on them every day to watch them change. What parts are you able to notice? What is the caterpillar doing?

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Do you recall in what stage of the life cycle the caterpillar is?

#### ★ TEACHER TIP

Raising butterflies provides many opportunities for object lessons on several stages of the insect's life over the next 2 weeks or so.

### WEEK 3 10m Natural History: Grade 2 - Lesson 3

#### *My Pet Spike*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

Tell me what you recall from our last reading. The caterpillar begins life as an egg and turns into a moth or butterfly. Today we're going to read about another caterpillar - up close!

#### → READ, NARRATE, & DISCUSS

p.11-16 "One summer" - "life cycle!"

- Last week we ordered egg -> caterpillar -> moth. Did you notice the step we missed in this week's reading? Draw the pupa and add it to your collection. Where does it go? (egg -> caterpillar/larva -> pupa -> adult/moth)

#### → SUPPLEMENTAL

∞ Video Link: Watch this caterpillar turn into a moth! (3:36)

#### ★ TEACHER TIP

Notice that any prompt to recall or think about can be done verbally, in picture, or in any other way that suits the student(s). Apply this flexibility to all lessons.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4 15m General Science: Grade 2 - Lesson 4

#### *Observing Caterpillars or Pupae Activity*

Materials: Nature Journal, caterpillars/pupae

PREP: Read Teacher Tip

#### → INTRO

#### ★ TEACHER TIP

The time that it takes for the butterflies to emerge will vary slightly, so enjoy them while you have them and release them based on the instructions that came with them.



## Term 1

Have you noticed if your caterpillars are changing? Take a close look today!

### → OBSERVE

Spend about 10 minutes observing your caterpillars or pupae today, and plan to check on them every day to watch them change. What do you notice today?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

## WEEK 4 10m Natural History: Grade 2 - Lesson 4

*How do caterpillars stay safe?*

Materials: Walks with a Naturalist 2

### → INTRO

Tell me what you remember about caterpillars. Look at the picture on p.18. What do you notice? Many birds love to eat caterpillars, so caterpillars need many ways to avoid being eaten! That's what we're going to learn about today.

### → READ, NARRATE, & DISCUSS

p.18-21 "I squeal" - "being eaten!"

- Think of a caterpillar you have seen. How was it trying to stay safe?

### → SUPPLEMENTAL

- ∞ Video Link: Caterpillars Hiding in Plain Sight (2:30)
- ∞ Video Link: Tips for finding caterpillars (3:43)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## WEEK 5 15m General Science: Grade 2 - Lesson 5

*Observing Pupae or Butterflies? Activity*

Materials: Nature Journal, pupae/butterflies

### → INTRO

How have your caterpillars changed this week? Take a close look today.

### → OBSERVE

Spend about 10 minutes observing your pupae or butterflies today, if you have not yet released them. What parts are you able to notice on the adult butterfly? What does the butterfly do when it emerges from the chrysalis? If you have released them already, then spend your time noticing wildflowers. What colors and shapes do you notice? Do ants, butterflies, or other insects visit them? If so, what do the insects look like? What are the insects doing?

### → NARRATE & DISCUSS

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.



## Term 1

Use a nature journal to record what the student(s) noticed.

- Tell about what you recorded.

### WEEK 5 10m Natural History: Grade 2 - Lesson 5

*Slug Caterpillars*

Materials: Walks with a Naturalist 2

#### → INTRO

Remind me what we read about last week. Look at the picture on p.23. What do you notice? Does that look like a caterpillar? Let's read about this very strange group of caterpillars.

#### → READ, NARRATE, & DISCUSS

p.23-26 "One of my" - "they eating?"

- Have you ever seen a strange caterpillar that might have been a slug caterpillar? Tell about it.
- Which of these caterpillars would you like to meet? What would you hope to observe?

#### → SUPPLEMENTAL

∞ Video Link: Caterpillars are the BEST! (9:27)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 6 15m General Science: Grade 2 - Lesson 6

*Planting Host Plants Activity*

Materials: resource link, seeds/plants for local butterfly species, pot with soil or place in the garden, trowel, water

PREP: Read Teacher Tip

#### → ACTIVITY

Plant one or more flowers for your local butterflies. You may plant seeds or seedlings, depending on what is most appropriate for your season and location.

#### ★ TEACHER TIP

To find out more about good host plants for your locale, ask your local nature preserve, conservancy, or state/national park, OR check out the resources at the link:

∞ Resource Link: Xerces Society

### WEEK 6 10m Natural History: Grade 2 - Lesson 6

*Tagging Monarchs*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

Tell me what you remember about caterpillars. Some caterpillars turn into moths, but others turn into butterflies. Look at the picture on p.27. What do you notice?

#### → READ, NARRATE, & DISCUSS

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).



## Term 1

p.27-31 "I see" - "amazing migration!"

- Compare the picture of the butterfly on p.27 with that of the moth on p.1. How are they similar or different?

### → SUPPLEMENTAL

∞ Video Link: How to Hold a Butterfly (4:16)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## WEEK 7 15m General Science: Grade 2 - Lesson 7

*Wildflower Watching Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Let's notice the wildflowers that are growing this season! Do you suppose that butterflies visit your wildflowers?

### → OBSERVE

Spend about 10 minutes noticing which wildflowers are growing this season. What colors and shapes do you notice? How many petals do they have? How are the petals arranged? Do ants, butterflies, or other insects visit them? If so, what do the insects look like? How many different types or sizes of insects? What are the insects doing?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

### ★ TEACHER TIP

Today is a good opportunity for an object lesson on wildflowers. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.

## WEEK 7 10m Natural History: Grade 2 - Lesson 7

*Acorn Weevils*

Materials: Walks with a Naturalist 2

PREP: Teacher Tip

### → INTRO

What do you remember from our last reading? Look at the picture of the acorn on p.33. What is going on with that acorn?

### → READ, NARRATE, & DISCUSS

p.33-35 "Our backyard" - "those holes!"

- Compare the stages in the life of the acorn weevil with those of the caterpillar. Are they the same? Do you think other insects follow this pattern? How could you find out?

### ★ TEACHER NOTE

It can sometimes be unnerving for students not to get an immediate answer to a question. Encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 1

### → SUPPLEMENTAL

- ∞ Video Link: Weevil Drilling an Acorn w/o narration (0:46)
- ∞ Video Link: This Weevil w/close up & narration (5:46)

### WEEK 8 15m General Science: Grade 2 - Lesson 8

#### *My Biome*

Materials: A Walk in the [familiar biome]

PREP: Read Teacher Tip

### → INTRO

Look at the picture on the cover. Do you recognize any of the living things on the cover? All of these (and more) live together in a community in our biome, so let's read about that today.

### → READ, NARRATE, & DISCUSS

A Walk in the [familiar biome] (whole book)

- What different habitats did you recognize in the book?

### ★ TEACHER TIP

Let students enjoy the photographs in this picture book. If you need to take a break and finish it another time, that is fine.

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 8 10m Natural History: Grade 2 - Lesson 8

#### *Me and My Magnifier*

Materials: Walks with a Naturalist 2

### → INTRO

What did we read about last week? Let's turn to the leaves of the trees today and see what we can learn.

### → READ, NARRATE, & DISCUSS

p.36-38 "I always carry" - "you notice?"

- Have you ever used a magnifier? What did you look at? What did you notice? What did it remind you of? What did you wonder about?

### → SUPPLEMENTAL

- ∞ Video Link: Leaves Magnified (3:59)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 9 15m General Science: Grade 2 - Lesson 9

#### *Observing a Community Activity*

Materials: Nature Journal

### → INTRO

Check on the wildflower that you planted for the butterflies.

### → OBSERVE

How has your wildflower grown or changed? Are there any insects or spiders visiting? Has it made seeds? Spend some time noticing who else lives in its habitat. What other plants are nearby? What birds do you see

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.



## Term 1

or hear?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell about what you recorded.

## WEEK 9 10m Natural History: Grade 2 - Lesson 9

*Red Oak, White Oak*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

### → INTRO

Recall what we read about last week. Today, Melissa is going to show us how much she notices when she compares 2 trees. You might want to make a drawing or diagram as we read.

### → READ, NARRATE, & DISCUSS

p.39-42 "Behind our house" - "is different?"

- Are there 2 trees that you would like to compare? Where are they?

### → SUPPLEMENTAL

∞ Video Link: Observation of Leaves (6:05)

### ★ TEACHER TIP

Because this passage is a description instead of a series of dependent events, building a diagram while reading can help to create a picture in the mind. If students need support, the teacher may do the activity with them in tandem or as a scribe.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## WEEK 10 15m General Science: Grade 2 - Lesson 10

*Noticing Insects Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Notice a fly or other insect today.

### → OBSERVE

Look at the insect closely with your magnifier. Describe its antennae, jaws, and eyes. How many wings does it have? From which sections do the wings come? The legs? What else do you notice?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell about what you found or what you recorded.

### ★ TEACHER TIP

Leave a small piece of meat outside in a safe location a few hours before the lesson. This can be a good opportunity for an object lesson with an insect, hopefully, a fly.

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.





## Term 1

### WEEK 10 10m Natural History: Grade 2 - Lesson 10

*How to Survive the Winter, Butterfly Style*

Materials: Walks with a Naturalist 2

#### → INTRO

Recall what we read last. Do you remember how the acorn weevil and Melissa's pet Spike spent the winter? The larva buries itself in the soil and spends the winter as a pupa, but just like different creatures have different ways of staying safe, different creatures have different ways of getting through the cold winter. Look at the picture on p.43. This butterfly is getting ready for winter, too. What do you notice?

#### → READ, NARRATE, & DISCUSS

p.43-45 "A few days" - "the winter?"

- Isn't it awesome how each creature has everything it needs in its home? Think of any creature you like and consider how it gets food, shelter, and safety. Read Matthew 6:26-27, if desired.

#### → SUPPLEMENTAL

- ∞ Video Link: Silvery Checkerspot Laying Eggs (2:34)
- ∞ Video Link: Monarch Butterfly Laying Eggs (0:52)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 11 15m General Science: Grade 2 - Lesson 11

*Revisiting Your Host Plant Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### → INTRO

Spend about 10 minutes with your plant from earlier in the term.

#### → OBSERVE

How has your plant grown or changed? Are there any animals visiting your plant yet? Does it have flowers? Has it made seeds? Describe any leaves or stalks.

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell about what you found or what you recorded.

#### ★ TEACHER TIP

If a student's particular plant will not sprout until next spring, then they might observe birds today instead.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 11 10m Natural History: Grade 2 - Lesson 11

*Catch Up*

#### → CATCH UP

Use this time to catch up on lessons, watch some videos that you didn't have time for, or discuss some interesting ideas over a snack. Depending

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

|                                                                                                                                                                                                                                                                                          |                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Term 1                                                                                                                                                                                                                                                                                   |                                                                                                               |
| on your discussions, these MIGHT include: <ul style="list-style-type: none"><li>• What do you recall about caterpillars?</li><li>• Tell about the stages in the life of many insects.</li><li>• Tell about leaves or nuts.</li><li>• Compare how different insects overwinter.</li></ul> |                                                                                                               |
| <div>WEEK 12 15m General Science: Grade 2 - Lesson 12</div> <div>Exam Week</div> <div>→ EXAMS</div> <div>Answer question(s) related to the course.</div> <div>Questions will come from:</div> <div>Nature Journal, Monarch and Milkweed, A Walk in the .....</div>                       | <div>• NATURE NOTEBOOK</div> <div>Prompt for General Science in Outdoor Work Quick Link.</div>                |
| <div>WEEK 12 10m Natural History: Grade 2 - Lesson 12</div> <div>Exam Week</div> <div>→ EXAMS</div> <div>Answer question(s) related to course.</div> <div>Questions will come from:</div> <div>Walks with a Naturalist 2</div>                                                           | <div>• OUTDOOR WORK</div> <div>Weekly walks and daily outdoor time are vital! Resources in Quick Links.</div> |



## Term 2

### WEEK 1 15m General Science: Grade 2 - Lesson 13

*Knowing a Tree Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### → INTRO

Let's spend some time to get to know a particular tree. This will be your tree, so pick one that you can revisit often.

#### → OBSERVE

Choose a special tree near your home or school (i.e., one that you can visit regularly). Spend about 10 minutes getting to know your tree. You can even give it a special name. Does it have leaves on it right now, or have the leaves fallen? Look at them closely, so that you can see them with your eyes closed. What color are the leaves? Is their shape oval, heart-shaped, star-shaped, or something else? Are the edges smooth or jagged/ridged? With your eyes closed, touch the bark with your fingertips. What do you notice? Do you think you could tell this tree's bark from another without looking? If not, touch it again to get to know it better.

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell about your tree.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

#### ★ TEACHER TIP

If students notice trees during their outdoor time, this could be a good opportunity for an object lesson. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.

### WEEK 1 10m Natural History: Grade 2 - Lesson 13

*A Monarch's Migration*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

Do you remember the caterpillars we learned about last term? Tell me about someone you remember. How do you think those creatures survive the winter?

#### → READ, NARRATE, & DISCUSS

p.47-52 "In 'How'" - "to or from?"

Read the notes and definitions at the end of each reading, as desired.

- Are you still noticing butterflies? If so, what were they doing? How do you think they are getting ready for winter?
- Are you noticing animals that are very busy or that haven't been around all summer? Tell what they have been doing.

#### → SUPPLEMENTAL

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 2

- ∞ Video Link: Monarch Butterfly Rest Stop (4:34)
- ∞ Video Link: Western Monarch Butterfly (3:51)
- ∞ Image Link: Monarch Migration Map

### WEEK 2 15m General Science: Grade 2 - Lesson 14

#### *Meeting a Keystone Species*

Materials: Desert Giant

PREP: Read Teacher Tip

#### → INTRO

Last term, you noticed that creatures in a community sometimes depend on each other, like butterflies and flowers. Have you ever noticed creatures other than butterflies and flowers that depend on each other? Look at the picture on the cover. This tree is a habitat for the birds, but so many other creatures depend on it, as well. So many creatures in its habitat depend on it that it is a "keystone species." Let's read and find out about that.

#### → READ, NARRATE, & DISCUSS

Desert Giant (whole book)

- Based on the story, what do you think is meant by "keystone species"? What creatures in your biome (or other creatures, if you are in the desert biome) seem to hold the whole community together?

#### → SUPPLEMENTAL

- ∞ Image Link: Wikipedia: Keystone

#### ★ TEACHER TIP

Some students will grasp the 'big picture' that a keystone holds the larger structure together, but others might need help or might get stuck on the word itself. Use the link or draw a diagram, if needed.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 2 10m Natural History: Grade 2 - Lesson 14

#### *Monarch Magic*

Materials: Walks with a Naturalist 2

#### → INTRO

Tell me what you remember about the monarch butterfly. Have you ever gone on a long trip, like the monarch? Melissa is going to take us to Mexico with her today, so that we can visit all of those monarchs that traveled for the winter.

#### → READ, NARRATE, & DISCUSS

p.53-56 "Last winter" - "magical moment!"

- Imagine what it would be like to be at a monarch overwintering site. How would you explore it with sight, hearing, touch, and smell?

#### → SUPPLEMENTAL

- ∞ Video Link: Monarch Butterfly Swarm (3:07)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 2

### WEEK 3 15m General Science: Grade 2 - Lesson 15

#### *Observing a Seashell Activity*

Materials: Nature Journal, seashells, optional: a rock

#### → INTRO

In the marine/saltwater biome, oysters are a keystone species. Oysters eat algae, balance nutrient levels in the water, provide food, and build habitats for other creatures. Like oysters, many ocean creatures live inside shells that give them a unique way of life. It is amazing to think that the shells we find at the beach are the homes of other creatures! Let's notice some of these beautiful creature homes today!

#### → OBSERVE

Spend about 10 minutes today noticing the beauty and uniqueness of one or more seashells. What colors and patterns do you notice? What is the texture? Where do you suppose the animal resides?

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell about what you recorded. Compare a seashell with a rock. How are they similar or different?

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 3 10m Natural History: Grade 2 - Lesson 15

#### *A Whale of a Trip*

Materials: Walks with a Naturalist 2

#### → INTRO

Remind me what we read last week. What else do you think we can see in Mexico? Look at the picture on p.57. Do you know what that is?

#### → READ, NARRATE, & DISCUSS

p.57-60 "In 'Monarch' - "nature journal!"

- What is the biggest animal you've ever seen? Where does it live?

#### → SUPPLEMENTAL

- ∞ Video Link: Humpback Whale Breaching (1:13)
- ∞ Video Link: What are barnacles? (0:55)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4 15m General Science: Grade 2 - Lesson 16

#### *Knowing a Tree Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### → INTRO

Visit your tree again to learn something new about your friend.

#### ★ TEACHER TIP

This could be a good opportunity for an object lesson on twigs.



## Term 2

### → OBSERVE

Notice any twigs and/or fruit that you can collect. Study the twigs and/or fruit for about 10 minutes. What color is the twig? How does the bark compare to that of the tree? Is it straight or are there any bumps or shoots? If so, how are those bumps/shoots arranged?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell what you learned about your tree.

## WEEK 4 10m Natural History: Grade 2 - Lesson 16

### Swash Riders

Materials: Walks with a Naturalist 2

### → INTRO

Tell me what you remember from our last lesson. Have you ever visited the ocean? If so, tell what you noticed. If not, tell what you would like to notice if you go there someday. Look at the picture on p.62. What do you notice about these shells?

### → READ, NARRATE, & DISCUSS

p.62-65 "When I visit" - "can observe!"

- A seashell is a home for an animal! How many different animals can you think of that carry their homes around with them? (clams, snails, hermit crabs, turtles, etc.) Would you like to carry your home around with you? Why or why not?

### → SUPPLEMENTAL

- ∞ Video Link: Coquina Clams (0:42)
- ∞ Video Link: How are Seashells Made? (4:28)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## WEEK 5 15m General Science: Grade 2 - Lesson 17

### Bird Watching Activity

Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Have you noticed which birds are visiting this week?

### → OBSERVE

Spend about 10 minutes bird watching. What colors, markings, and shapes do you notice? Do they visit particular plants or trees? What is the bird doing? What is its posture (i.e., how does it hold its body)?

### → NARRATE & DISCUSS

### ★ TEACHER TIP

This could be a good opportunity for an object lesson on birds.



## Term 2

Use a nature journal to record what the student(s) noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

### WEEK 5 10m Natural History: Grade 2 - Lesson 17

#### *Beach Treasures*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

Tell me what you recall from our last reading. What else can you find on the beach? Look at the picture on p.66. What do you notice?

#### → READ, NARRATE, & DISCUSS

p.66-70 "Taking a walk" - "nature walk?"

- Tell about your favorite place to walk - a place that seems filled with treasures.

#### → SUPPLEMENTAL

∞ Video Link: Sand Dollar (4:22)

∞ Video Link: Stingrays & Skates (0:23)

∞ Video Link: Jellies 101 (2:47)

#### ★ TEACHER TIP

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#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 6 15m General Science: Grade 2 - Lesson 18

#### *Knowing a Tree Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### → INTRO

Sometimes we notice more when we compare. Let's try this today to notice even more about our tree.

#### → OBSERVE

Collect twigs from your tree and a different tree, and spend about 10 minutes studying them. What color is the twig? How does the bark compare to that of the tree? Is it straight or are there any bumps or shoots? If so, how are those bumps/shoots arranged?

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell what you learned about your tree.

#### ★ TEACHER TIP

This could be a good opportunity for an object lesson on twigs. Questions are embedded in the lesson to help, as needed. If preferred, teachers can refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.



## Term 2

### WEEK 6 10m Natural History: Grade 2 - Lesson 18

#### *Tracks in the Snow*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

Recall what we've been reading about. Now it's time to leave the ocean and explore the snowy parts of our home. Do you get snow where you live? Look at the picture on p.71. What do you notice in this picture? Have you seen wild animals that live in snowy habitats? Let's go looking for one today.

#### → READ, NARRATE, & DISCUSS

p.71-74 "Back in North" - "muddy areas!"

- Sometimes when we're on a walk or working on a puzzle, there seems to be a mess of pieces, and it's hard to see how they fit together. Have you ever experienced that? Do you think that's how Melissa felt when she was trying to understand the tracks through the snow? What do you do in such a situation?

#### → SUPPLEMENTAL

∞ Video Link: Ermine in the Snow (0:10)

∞ Video Link: Track Patterns (3:38)

#### ★ TEACHER TIP

It can sometimes be unnerving for students to not get an immediate answer to a question, so encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 7 15m General Science: Grade 2 - Lesson 19

#### *Birds and Their Nests*

Materials: Mama Built a Little Nest

PREP: Read Teacher Tip

#### → INTRO

Have you ever found a bird nest? What was it made of? How was it made? Where was it? What trees or other plants were nearby? These are all clues to who built it because each bird lives differently within its community.

#### → READ, NARRATE, & DISCUSS

Mama Built a Little Nest (whole book)

- Do you think anyone has had a nest in your tree? Why or why not?

#### ★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.





## Term 2

### WEEK 7 10m Natural History: Grade 2 - Lesson 19

*Antlers versus Horns*

Materials: Walks with a Naturalist 2

#### → INTRO

Remind me what we read about last time. Tracks are one way that we can learn about a creature, even if we don't get to see it. Can you think of other signs that animals might leave behind after they've left? (scat, remains of food, nesting material, antlers, etc.) Look at the picture on p.76. What are those? What do you notice?

#### → READ, NARRATE, & DISCUSS

p.76-79 "When my nephew" - "in the woods?"

- What animals have antlers or horns where you live?

#### → SUPPLEMENTAL

- ∞ Video Link: North America's Deer (0:52)
- ∞ Video Link: Bull Moose Shedding its Antlers (1:18)
- ∞ Video Link: Bighorn Sheep (0:12)
- ∞ Video Link: Mountain Goat (0:59)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 8 15m General Science: Grade 2 - Lesson 20

*Knowing a Tree Activity*

Materials: Nature Journal

#### → INTRO

Changing our perspective (our view) can also help us notice more. Let's change our perspective and see what else we can notice about our tree.

#### → OBSERVE

Observe your tree from a distance and up close. How is your tree shaped? What is the land like around it? Do other plants or animals live near your tree? How do you think they help your tree? How do you suppose your tree helps them?

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell what you learned about your tree.

### WEEK 8 10m Natural History: Grade 2 - Lesson 20

*A Plant that Bites Back*

Materials: Walks with a Naturalist 2

#### → INTRO

We've been visiting some different places and learning about different creatures this term. Tell me about one or two that you remember. One

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 2

place that a lot of people don't think about visiting is a swamp. Have you ever been to a swamp? What do you think of when you hear that word? Let's visit a swamp with Melissa today.

### → READ, NARRATE, & DISCUSS

p.80-82 "I like to" - "can discover!"

- Think about a plant that you enjoy. Tell what makes that plant friend interesting to you!

### → SUPPLEMENTAL

∞ Video Link: Venus Fly Traps Can Count (1:44)

∞ Website Link: Interesting Native Plants. Most plants have something interesting once you get to know them, but here are a few images with short descriptions for inspiration.

## WEEK 9 15m General Science: Grade 2 - Lesson 21

*Nest Investigation Activity*

📋 Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Spend about 10 minutes today looking for or studying a bird nest. If you don't find one, that's still a good observation! Note where you looked and what you did see during your search.

### → OBSERVE

Since the songbirds are not nesting in the winter, any nests you find should be old and not in use, but observe them in place anyway. Another creature may reuse the nest or its material in the future. Discover as many clues as you can as to who built the nest. What was it made of? How was it made? Where was it? What trees or other plants were nearby?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- If you found a nest, do you have an idea about who built it? You can ask a local naturalist or park ranger to confirm your guess - they love to help!

### ★ TEACHER TIP

This could be a good opportunity for an object lesson on a bird's nest.

## WEEK 9 10m Natural History: Grade 2 - Lesson 21

*Night in a Blind*

📋 Materials: Walks with a Naturalist 2

### → INTRO

Tell what you remember from our last reading. Look at the picture on p.84. Have you ever seen a bird like this? What do you notice about the bird?

### ★ TEACHER NOTE

**There is an old earth reference at 2:48** in the video. Teachers might choose to allow it to pass by the way, provide additional commentary, or skip this one.

### • OUTDOOR WORK

Weekly walks and daily outdoor



## Term 2

### → READ, NARRATE, & DISCUSS

p.84-88 "One March" - "never forget!"

- Do the birds at home act differently when they can see you and when they don't know you are there? Tell what you've noticed.

### → SUPPLEMENTAL

∞ Video Link: Sandhill Cranes (4:56)

∞ Image Link: Simple Bird Blind

time are vital! Resources in Quick Links.

## WEEK 10 15m General Science: Grade 2 - Lesson 22

*Nest Investigation Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Spend about 10 minutes today looking for or studying a bird nest. If you didn't find one last week, look in a different location or in different places today. You are a nature detective!

### → OBSERVE

Since the songbirds are not nesting in the winter, any nests you find should be old and not in use, but observe them in place anyway. Another creature may reuse the nest or its material in the future. Discover as many clues as you can as to who built the nest. What was it made of? How was it made? Where was it? What trees or other plants were nearby?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- If you found a nest, do you have an idea about who built it? You can ask a local naturalist or park ranger to confirm your guess - they love to help!

### ★ TEACHER TIP

This could be a good opportunity for an object lesson on a bird's nest.

## WEEK 10 10m Natural History: Grade 2 - Lesson 22

*A Swamp Paddle*

Materials: Walks with a Naturalist 2

### → INTRO

Recall what we read about last week. Today let's see what other creatures we can find in Melissa's canoe.

### → READ, NARRATE, & DISCUSS

p.89-94 "In April" - "you all?"

- Tell about a place you have enjoyed exploring. What new nature friends did you meet there?

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 2

### → SUPPLEMENTAL

∞ Video Link: Mingo Swamp, Missouri (6:34)

#### WEEK 11 15m General Science: Grade 2 - Lesson 23

##### *Bird Watching Activity*

📋 Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Which birds are visiting this week? Let's take notice!

### → OBSERVE

Spend about 10 minutes bird watching. What colors, markings, and shapes do you notice? Do they visit particular plants or trees? What is the bird doing? What is its posture (i.e., how does it hold its body)?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

### ★ TEACHER TIP

This could be a good opportunity for an object lesson on a bird.

#### WEEK 11 10m Natural History: Grade 2 - Lesson 23

##### *Catch Up*

### → CATCH UP

Use this time to catch up on lessons, watch some videos that you didn't have time for, or discuss some interesting ideas over a snack. Depending on your discussions, these MIGHT include:

- Recall different ways that moths and butterflies overwinter (as egg, pupa, migrating adult).
- Recall different places we 'visited' (Mexico, seaside, swamp).
- Tell about the variety of different creatures at the seaside.
- Consider the different signs that animals leave behind and that help us learn about them (tracks, antlers, their homes, etc).
- Recall the Things we treasure and nature friends we've met.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

#### WEEK 12 15m General Science: Grade 2 - Lesson 24

##### *Exam Week*

### → EXAMS

Answer question(s) related to the course.

Questions will come from:

Nature Journal, Mama Built a Little Nest, Desert Giant



## Term 2

**WEEK 12** 10m Natural History: Grade 2 - Lesson 24

*Exam Week*

**→ EXAMS**

Answer question(s) related to course.

Questions will come from:  
Walks with a Naturalist 2

**• OUTDOOR WORK**

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### WEEK 1 15m General Science: Grade 2 - Lesson 25

*Meeting a Unique Mammal*

Materials: The Secret Life of the Skunk

PREP: Read Teacher Tip

#### → INTRO

Look at the picture on the cover. Have you ever seen a skunk? What would you do if you came across one on a walk? We don't often get to learn about skunks or any of the wild mammals in our biome up close, so let's read to find out more.

#### → READ, NARRATE, & DISCUSS

The Secret Life of the Skunk (whole book)

- Have you ever smelled a skunk before? Take note of any skunky smell this term. When and where does it occur? How long does it last? Foxes smell very similar to skunks during mating season, but their season is a little earlier than skunks, and their odor doesn't tend to 'stick' quite as long.

#### ★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 1 10m Natural History: Grade 2 - Lesson 25

*The Bear Woods*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

Do you remember when Melissa tracked the ermine in the snow last term? Tell me what you remember. Spring is a really great time to notice animal tracks. Why do you think that is? (mud, animals becoming more active, etc.) Let's listen to find out about many different ways we can notice animals even when we might not get to see them.

#### → READ, NARRATE, & DISCUSS

p.95-99 "Mike and I" - "leave behind!"

- Do you leave signs behind when you go outside? What kinds of signs do you think you leave? Have you ever walked in the mud? What did your footprints look like?

#### → SUPPLEMENTAL

∞ Video Link: Studying Individual Tracks (3:39)

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### WEEK 2 15m General Science: Grade 2 - Lesson 26

#### *Mammal Investigation Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### → INTRO

Wild mammals usually hide from humans, but we can learn about them by following their tracks. Spots with soft or wet soil are good for footprints, but if your soil is very dry, then you might still be able to find other signs, like scat or leftover food.

#### → OBSERVE

Spend about 10 minutes today looking for or studying animal tracks (or other sign). What is the size and shape? Are there claws visible? How many toes? Is there more than one animal? Are they the same or different? Which way are they pointing? How would you describe this habitat? What other plants and animals live in this biome?

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Do you know who made any tracks or left any sign that you found? If not, what other clues might help? If so, you can ask a local naturalist or park ranger to confirm your guess - they love to help!

#### ★ TEACHER TIP

This could be a good opportunity for an object lesson on animal signs. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.

### WEEK 2 10m Natural History: Grade 2 - Lesson 26

#### *Barking up the Right Tree*

Materials: Walks with a Naturalist 2

#### → INTRO

Recall what we read last time. There is a very interesting creature that you might observe when you are outside. It often grows on trees and rocks. Have you ever heard of lichen? Let's find out about this strange nature find today.

#### → READ, NARRATE, & DISCUSS

p.100-102 "There are lots" - "your magnifier?"

- What do you notice about the lichen in the picture on p.101? What do you wonder about? Does it remind you of anything?

#### → SUPPLEMENTAL

∞ Website Link: Lichens Photo Gallery

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### WEEK 3 15m General Science: Grade 2 - Lesson 27

*Knowing a Tree Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### → INTRO

Revisit your tree today to see what has changed.

#### → OBSERVE

Closely examine its twigs and look carefully at the buds. What color is the twig? What color and shape are the buds? Study the form of the buds. Can you guess what the layers of the bud will look like when they open? How are those buds arranged? Bring a small twig inside to place in some water. Watch the buds throughout the day to observe any changes.

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell what you learned about your tree.

#### ★ TEACHER TIP

This could be a good opportunity for an object lesson on a twig.

### WEEK 3 10m Natural History: Grade 2 - Lesson 27

*Plant Time Machine*

Materials: Walks with a Naturalist 2

#### → INTRO

Remind me what we've been reading about. Today we're going to be thinking about the plants that are starting to grow and how we can get to know them better. Look at the picture on p.104. What do you notice in this picture?

#### → READ, NARRATE, & DISCUSS

p.104-108 "In the spring" - "grow and change?"

- What plants have you been noticing outside? Are they growing? Have you noticed similar plants with and without flowers? Tell what you have seen.

#### → SUPPLEMENTAL

∞ Video Link: Yellow Trout Lily (2:22)

∞ Website Link: Gallery of Fawn (Trout) Lilies Across North America

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4 15m General Science: Grade 2 - Lesson 28

*Bird Watching Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

#### ★ TEACHER TIP

This could be a good opportunity for an object lesson on a bird.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.





## Term 3

### → INTRO

Today is a good day to notice which birds are visiting.

### → OBSERVE

Spend about 10 minutes bird watching. What colors, markings, and shapes do you notice? Do they visit particular plants or trees? Do they visit particular plants or trees? If so, study the tree to see if you can find reasons for this. What is the bird doing?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Do you notice anyone already familiar to you? Do you notice anyone new?

## WEEK 4 10m Natural History: Grade 2 - Lesson 28

*A Bee Mystery*

Materials: Walks with a Naturalist 2

### → INTRO

Remind me what we read about last time. Look at the picture on p.110. Tell what you notice about this creature. Is it an insect? How do you know?

### → READ, NARRATE, & DISCUSS

p.110-113 "One afternoon" - "can observe!"

- How do you feel about bees when you see them on a walk or when you play outside? Do they seem to notice you, or do they seem to be doing their business?

### → SUPPLEMENTAL

∞ Video Link: Cellophane Bees (1:26)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## WEEK 5 15m General Science: Grade 2 - Lesson 29

*Looking for Beetles Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Today, you will look for beetles, an insect that has a pair of hardened wings covering their flight wings.

### → OBSERVE

Some good places to look are under small rocks, on plants, and in grass and leaf litter. Use a stick to safely lift rocks. If you find beetles, take some time to observe them. Use a magnifier if you have one. What parts of the body do you see? What colors are its hard shell? Describe its antennae, jaws, and eyes. If you give it a gentle nudge with a blade of grass or leaf,

### ★ TEACHER TIP

This could be a good opportunity for an object lesson on a beetle or other insect. Questions are embedded in the lesson to help, as needed. If preferred, teachers can refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.



## Term 3

does it take off and let you see its wings? If you find someone else that you want to observe, you may take the time to do so. Spend about 10 minutes looking for beetles.

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Did you find beetles? If so, describe their habitat. How does it provide food and shelter for them?

### → SUPPLEMENTAL

∞ Video Link: Ladybug Wings (1:26)

## WEEK 5 10m Natural History: Grade 2 - Lesson 29

### *Bird Banding Part 1*

📄 Materials: Walks with a Naturalist 2

### → INTRO

We have been learning about many signs of spring. Tell me about one or two that you remember. What do you recall about the cellophane bee? Look at the picture on p.115. What do you notice about this bird?

### → READ, NARRATE, & DISCUSS

p.115-118 "When I first" - "banding station."

- Have you ever held a bird before? If so, what did it feel like? What did you notice? If not, tell about a bird that you would like to know better.

### → SUPPLEMENTAL

∞ Video Link: Bird Banding (to 1:20)

∞ Video Link: Hooded Warbler (0:11)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## WEEK 6 15m General Science: Grade 2 - Lesson 30

### *Looking for Beetles Activity*

📄 Materials: Nature Journal

### → INTRO

Today, you will look for beetles again. If you find beetles, take some time to observe them. Use a magnifier if you have one. If not, then observe whatever nature provides.

### → OBSERVE

If you find a beetle, notice what parts of the body you see. What colors are its hard shell? Describe its antennae, jaws, and eyes. If you give it a gentle nudge with a blade of grass or leaf, does it take off and let you see its wings? If you find someone else that you want to observe, you may take the time to do so. Spend about 10 minutes looking for beetles.

### → NARRATE & DISCUSS

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.



## Term 3

Use a nature journal to record what the student(s) noticed.

- Did you find beetles? If so, describe their habitat. How does it provide food and shelter for them?

### WEEK 6 10m Natural History: Grade 2 - Lesson 30

*Bird Banding Part 2*

Materials: Walks with a Naturalist 2

PREP: Read Teacher Tip

#### → INTRO

In the last passage, we learned a bit about bird banding. Tell what you remember. Let's listen to the rest of this story.

#### → READ, NARRATE, & DISCUSS

p.119-123 "At the banding" - "your bird!"

- Tell about birds that you have noticed this week.

#### → SUPPLEMENTAL

∞ Video Link: Bird Banding (1:21-3:22)

#### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 7 15m General Science: Grade 2 - Lesson 31

*Your Biome*

Materials: A Walk in the [different biome]

PREP: Read Teacher Tip

#### → INTRO

Look at the picture on the cover. Do you recognize any of the living things on the cover? This book is about a living community or biome that is different from ours, so let's read and find out the ways that it is similar, but unique.

#### → READ, NARRATE, & DISCUSS

A Walk in the [different biome] (whole book)

- Did you recognize habitats in this book that are similar to some where you live? What plants and animals were the same or similar and which were different? Were there habitats or creatures (e.g. insects, birds, mammals, trees, etc) in them that were very different from anything where you live?

#### ★ TEACHER TIP

Let students enjoy the photographs in this picture book. If you need to take a break and finish it another time, that is fine.



## Term 3

### WEEK 7 10m Natural History: Grade 2 - Lesson 31

*Froggy Lullaby*

Materials: Walks with a Naturalist 2

#### → INTRO

Tell me about our last reading. Look at the picture on p.124. What do you notice? What do you wonder about?

#### → READ, NARRATE, & DISCUSS

p.124-127 "At our house" - "nighttime sounds!"

- Can you sing like a frog that you have heard? Give it a try!

#### → SUPPLEMENTAL

∞ Video Link: Some Frog Calls (1:24)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 8 15m General Science: Grade 2 - Lesson 32

*Comparing Tracks Activity*

Materials: Nature Journal

#### → INTRO

What kinds of tracks have you noticed? Let's compare different tracks today and think about what stands out.

#### → OBSERVE

Spend about 10 minutes studying and copying the tracks at the link (or some of them). Which ones stand out as being unique? Which ones might be easily confused? What features are distinctive about each one?

∞ Image Link: Track and Sign

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Have you seen any of these tracks before?

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 8 10m Natural History: Grade 2 - Lesson 32

*Dragonfly Swarm*

Materials: Walks with a Naturalist 2

#### → INTRO

Tell me what you remember from our last reading. Look at the picture on p.128. What do you notice? Today, we're noticing the dragonfly, an insect we might see living alongside frogs and other creatures that like wet habitats.

#### → READ, NARRATE, & DISCUSS

p.128-131 "One day - "they're up to!"

#### ★ TEACHER NOTE

The supplemental video mentions the age of the Earth at 3:34. Teachers might want to allow it to pass by the way, consider how they would like to discuss, stop the video early, or skip this one.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

- If you were a superhero called The Dragonfly, what would be your superpower?

### → SUPPLEMENTAL

∞ Video Link: How do dragonflies see the world? (4:13)

#### WEEK 9 15m General Science: Grade 2 - Lesson 33

*Mammal Investigation Activity*

📖 Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Spend some time studying and comparing any tracks that you have recorded so far, or look for more in the soft earth.

### → OBSERVE

Do any of your tracks have the distinctive features you noticed last week? What is the size and shape? Are there claws visible? How many toes? What is the land like near the animal sign? Are there other signs of animal activity?

### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Do you know who made the tracks or left the signs that you found? If so, you can ask a local naturalist or park ranger to confirm your guess - they love to help!

#### ★ TEACHER TIP

This could be a good opportunity for an object lesson on animal signs.

#### WEEK 9 10m Natural History: Grade 2 - Lesson 33

*Dobsonflies and Hellgrammites*

📖 Materials: Walks with a Naturalist 2

### → INTRO

Tell me what you recall from our last reading. Look at the picture on p.132. What do you notice? What do you think this might be?

### → READ, NARRATE, & DISCUSS

p.132-136 "On a walk" - "surprise you?"

- Do big insects scare you when you are outside? Do you think that learning about them helps you feel less scared? Is there anything else that helps?

### → SUPPLEMENTAL

∞ Video Link: Male Dobsonfly (1:47)

∞ Video Link: Hellgrammites (1:08)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### WEEK 10 15m General Science: Grade 2 - Lesson 34

*Knowing a Tree Activity*

Materials: Nature Journal

#### → INTRO

Revisit your tree again this week.

#### → OBSERVE

Closely examine your tree's twigs and look carefully at the buds if they remain. If they have already opened, look carefully at the new flowers and/or leaves. What is their color and texture? How many are grouped together? What is their shape? Do all of the flowers on your tree look the same, or are there different kinds of flowers on the same tree?

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell what you learned about your tree.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 10 10m Natural History: Grade 2 - Lesson 34

*Flying Blue Toothpicks*

Materials: Walks with a Naturalist 2

#### → INTRO

Remind me what we read about in our last reading. What do you remember about the dragonflies from the previous week? The larvae of dragonflies and their cousins, damselflies, live in the water, too. Look at the picture on p.137. What do you notice about the damselfly larva? Today we'll read about damselflies and their life cycle.

#### → READ, NARRATE, & DISCUSS

p.137-139 "When I was" - "an exuvia!"

- Compare the life cycle of damselflies to that of butterflies or dobsonflies. What stages does the damselfly have in its life cycle? In what sequence do they occur? (egg -> larva/nymph -> adult) Dragonflies have a similar life cycle to the damselfly. What do you recall about the butterfly or dobsonfly? (egg -> larva -> pupa -> adult)

#### → SUPPLEMENTAL

- ∞ Video Link: Damselflies (2:31)
- ∞ Video Link: Damselfly Emergence (1:28)
- ∞ Video Link: Dragonfly or Damselfly? (0:56)
- ∞ Video Link: Green Lacewing (0:32)

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### WEEK 11 ☐ 15m General Science: Grade 2 - Lesson 35

#### *Noticing Insects Activity*

☐ Materials: Nature Journal

#### → INTRO

Notice flying insects today.

#### → OBSERVE

Are the bees active? Have you seen butterflies or beetles flying? Are you close enough to the water to see dragonflies, damselflies, or lacewings? What do you notice these insects doing? Where are they? Can you observe closely enough to examine their wings? Compare how different insects fly. How do they use their wings differently?

#### → NARRATE & DISCUSS

Use a nature journal to record what the student(s) noticed.

- Tell about what you found or what you recorded.

### WEEK 11 ☐ 10m Natural History: Grade 2 - Lesson 35

#### *Catch Up*

#### → CATCH UP

Use this time to catch up on lessons, watch some videos that you didn't have time for, or discuss some interesting ideas over a snack. Depending on your discussions, these MIGHT include:

- Consider the different signs that animals leave behind and that help us learn about them (tracks, scat, their homes, etc).
- Think about what kinds of observations are helpful to notice about individual tracks.
- Tell about signs of spring (birds, plants in various stages of growth, native bees and other early insects, etc).
- Tell what you recall about lichen.
- Tell about frogs and pond insects.
- Compare the life cycle of damselflies or dragonflies and that of butterflies or dobsonflies.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 12 ☐ 15m General Science: Grade 2 - Lesson 36

#### *Exam Week*

#### → EXAMS

Answer question(s) related to the course.

Questions will come from:

Nature Journal, A Walk in the ...., The Secret Life of the Skunk



## Term 3

WEEK 12 10m Natural History: Grade 2 - Lesson 36

*Exam Week*

→ EXAMS

Answer question(s) related to course.

Questions will come from:  
Walks with a Naturalist 2

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



# Science: Grade 2

## Examination

### Term 1

#### General Science: Grade 2

- Tell about or draw 3 different habitats in your biome in which a butterfly lives.

#### Natural History: Grade 2

- Walks with a Naturalist 2: Tell about or draw a caterpillar. OR Tell about or draw a leaf or tree nut.

### Term 2

#### General Science: Grade 2

- Tell about or draw a tree in your biome with at least 3 ways that others depend upon it.

#### Natural History: Grade 2

- Walks with a Naturalist 2: Tell about or draw the ways that insects overwinter. OR Tell about or draw a creature that lives in an unfamiliar habitat, such as the seaside or swamp.

### Term 3

#### General Science: Grade 2

- Tell about or draw a creature from your biome and compare it to a creature from another biome.

#### Natural History: Grade 2

- Walks with a Naturalist 2: Tell about or draw signs that animals leave behind. OR Compare using words or pictures the life cycle of a moth with that of the damselfly.